

Phase 3 (Extended Reach) Subtask Deepening — epic → task → subtask (closes R5)

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Volume 7 (Phase Execution), document 5 of 5. Companion to subtask-deepening-p1.md / -p2.md, applying the R5 deepening (REFINEMENT_NOTES.md) to Phase 3: every task HVPN-P3-NNN in 09-phase3-reach-wbs.md is decomposed into PR-sized subtasks HVPN-P3-NNN.k (§11.4.93/.54), each with a concrete **acceptance** (falsifiable, captured-evidence §11.4.5/.69/.107), the **§11.4.169 test types** (the 09-... §1 vocabulary incl. the Phase-3-specific REPRO), and an **estimated complexity** (engineer-day T-shirt, **widest error bars in the programme**, sizing-only TARGET, §11.4.6). **Spec-only.** Phase 3 is the one phase where *"ship what's provable, flag the rest honestly"* is the correct stance (09-... §13): the two native shims (E21 HarmonyOS, E22 Aurora) are real native work on toolchains the project does not yet own hardware/CI for, so device-gated acceptances are marked **PENDING_DEVICE:** (§11.4.3 hardware_not_present) and engagement-gated ones **Operator-blocked** (§11.4.21/.101) — never a faked pass. Every subtask traces to its parent task in 09-.... These .k rows feed the workable-items DB (workable-items-model.md §7).

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0. Deepening conventions + the honesty stance

Columns as in -p1/-p2 §0: id (HVPN-P3-NNN.k) · **Subtask** (≥ 6 words §11.4.91) · **Acceptance** · **Tests** (§11.4.169 codes, 09-... §1 vocab incl. REPRO) · **Cx** (XS/S/M/L engineer-days, sizing TARGET).

Phase-3 honesty stance (§11.4.6, 09-... §0/§13). Where a subtask's acceptance genuinely cannot be proven without the target device or an external engagement, the acceptance carries the literal PENDING_DEVICE: (device-gated, §11.4.3 hardware_not_present) or the item's status is Operator-blocked (engagement / spend, §11.4.21/.101) with the §11.4.148 D3 unblock condition. A gate is **never** marked pass on metadata/config-only evidence; the runtime signature (§11.4.108) for the native shims is asserted on a **clean install on the real device** — an APK/HAP/RPM that *builds* is not *done*. No active CI (§11.4.156): the "fork runners" are local/self-hosted rootless-Podman build pods, never GitHub/GitLab workflows.

1. E20 — Reach CI fabric (gate G20)

Self-hosted rootless Podman build pods (§11.4.76/.161/.156); a fork lag never blocks mainline.

HVPN-P3-200 — Pin OpenHarmony-SIG Flutter fork + DevEco toolchain in a pod (09-... §5; M)

id	Subtask	Acceptance	Tests	Cx
.1	harmonyos.container quadlet building a HAP from the pinned flutter_flutter ohos SHA	podman build produces a HAP from the sample app; toolchain SHA recorded	UNIT,FA	M
.2	DevEco signing material mounted read-only as secrets (§11.4.10)	secrets never in the image layer (SEC leak audit green)	SEC	S
.3	REPRO: same SHA → same toolchain digest	identical toolchain digest on rebuild (REPRO)	REPRO	S

HVPN-P3-201 — Pin OMP Aurora flutter-aurora fork + RPM signing pod (09-... §5; M)

id	Subtask	Acceptance	Tests	Cx
.1	aurora.container quadlet producing a signed Aurora RPM	signed RPM of the sample app builds;	UNIT,FA	M

id	Subtask	Acceptance	Tests	Cx
.2	Mirror-pin the Russian-hosted SDK (§11.4.77 regen) + isolated egress allow-list	rpm -K verifies the signature SDK tarball SHA-256 matches the recorded manifest; egress allow-list holds	SEC,REPRO	S

HVPN-P3-202 — Reach release-artifact registry + provenance manifest (09-... §5; S)

id	Subtask	Acceptance	Tests	Cx
.1	Signed reach_artifacts.json (source SHA + toolchain digest + SBOM pointer per HAP/RPM/.wasm)	manifest entry per build, schema-validated, links resolve	UNIT,FA	S

HVPN-P3-203 — G20 readiness certification (09-... §5; XS)

id	Subtask	Acceptance	Tests	Cx
.1	make reach-ci-gate builds+signs both fork artefacts, emits the G20 verdict	HAP + signed RPM both produced from a clean checkout in two -count=2 runs; gate outcome=pass	FA,CHAL	XS

2. E21 — HarmonyOS NEXT build (G21, highest risk)

ArkTS VPN ability + NAPI bridge to the Rust helix-core .so; the UI ports for free, the tunnel ability is real native work.

HVPN-P3-210 — helix-core cross-compile to OpenHarmony + NAPI surface (09-... §6; L)

id	Subtask	Acceptance	Tests	Cx
.1	Build helix-ffi for aarch64-unknown-linux-ohos; emit libhelix_ohos.so NAPI module (re-exports the platform-neutral core, §11.4.28)	.so loads in an ArkTS test harness; helix_version() round-trips	UNIT,INT	M
.2	helix_start(fd,cfg)/helix_stop/ helix_subscribe NAPI surface over the unchanged core	NAPI surface callable from ArkTS (INT)	UNIT,INT	M

id	Subtask	Acceptance	Tests	Cx
.3	MEM/BENCH on emulator	RSS ceiling sampled (PENDING_DEVICE: on hardware); throughput benched on emulator	MEM,BENCH	M

HVPN-P3-211 — Network Kit VpnExtensionAbility + lifecycle + kill-switch
(09–... §6; L)

id	Subtask	Acceptance	Tests	Cx
.1	ArkTS ability: VPN consent → VpnConnection (tun fd/routes/ DNS, MTU 1280) → hand fd to core	ability creates the tunnel fd + drives the core (INT)	UNIT,INT	M
.2	Drive connect/ disconnect/kill-switch/auto-escalate from the core status stream	PENDING_DEVICE: enroll→UP→curl reaches authorized LAN host on a HarmonyOS NEXT device; window-scoped MP4 + DevEco heap capture; §11.4.3 SKIP hardware_not_present until G20 provisions a device	E2E,MEM,UX,REC	L
.3	Tunnel-drop blanks plaintext (kill-switch per §04)	PENDING_DEVICE: kill-switch blanks plaintext on drop (device capture)	SEC,CHAL	M

HVPN-P3-212 — Flutter HAP build (Access + Connector, OHOS channel)
(09–... §6; M)

id	Subtask	Acceptance	Tests	Cx
.1	Signed HAPs via runHelixApp(flavor,...) on the SIG fork; ArkTS[?]Dart MethodChannel wired	app installs + drives the shim; golden UI tests pass on the OHOS theme	UI	M
.2	UX walkthrough MP4 (PENDING_DEVICE)	UX MP4 vision-verified (§11.4.159), PENDING_DEVICE: until a device	UX,REC	S

HVPN-P3-213 — DevEco signing + market metadata + l10n (zh-Hans first-tier)
(09–... §6; S)

id	Subtask	Acceptance	Tests	Cx
.1	DevEco signing profile + AppGallery metadata + intl_zh.arb first-tier locale	signed HAP verifies; zh-Hans strings render with no overflow (UI golden)	FA,UI	S

HVPN-P3-214 — G21 device certification + honest gap doc (09-... §6; M)

id	Subtask	Acceptance	Tests	Cx
.1	challenges/helix_qa bank: enroll→UP→reach→kill-switch on device, PASS only on captured evidence (§11.4.27/.107)	gate outcome=pass with device evidence, OR outcome=pending_device with the §11.4.148 D3 unblock condition (provision a HarmonyOS NEXT device + add to the §11.4.128 tracked-device set) — never a faked pass	E2E,FA,CHAL,REC	M

3. E22 — Aurora OS build (G22)

OMP Russia Flutter fork → signed RPM; tunnel backend in Qt/C++ linking helix-core as a C library. Enterprise/government SKU.

HVPN-P3-220 — helix-core C ABI for Aurora + cbindgen header (09-... §7; L)

id	Subtask	Acceptance	Tests	Cx
.1	Build helix-ffi for the aarch64/armv7hl Sailfish target; cbindgen → helix.h (no Aurora specifics inside the core, §11.4.28)	header compiles in a Qt test; helix_version() round-trips	UNIT,INT	M
.2	MEM/BENCH on the Aurora target	RSS sampled (PENDING_DEVICE:); throughput benched	MEM,BENCH	M

HVPN-P3-221 — Qt/C++ tun lifecycle + Friflex Flutter bridge + kill-switch (09-... §7; L)

id	Subtask	Acceptance	Tests	Cx
.1	Open/route/DNS the Sailfish tun; helix_start over the C ABI; status marshalled onto	the backend drives the core; status events surface (INT)	UNIT,INT	M

id	Subtask	Acceptance	Tests	Cx
	the Qt event loop			
. 2	Friflex-style plugin bridge connect/disconnect/status to Flutter + kill-switch/auto-escalate	PENDING_DEVICE: enroll→UP→reach on an Aurora device; kill-switch blanks plaintext; MP4 + RSS capture; §11.4.3 SKIP until a device	E2E, MEM, UX, REC, CHAL	L

HVPN-P3-222 — Flutter signed-RPM build (Access + Connector, OMP fork)
(09–... §7; M)

id	Subtask	Acceptance	Tests	Cx
. 1	Signed Aurora RPMs of the two flavors + ru first-tier 110n	rpm -K verifies; ru strings render with no overflow; install on the Aurora emulator (PENDING_DEVICE: for hardware reach)	UI, UX, REC, SEC	M

HVPN-P3-223 — G22 device certification + enterprise-SKU ops doc (09–... §7; M)

id	Subtask	Acceptance	Tests	Cx
. 1	Bank entry enroll→UP→reach→kill-switch on Aurora + ops doc (isolated runner + Mos.Hub signing §11.4.10)	gate outcome=pass with device evidence, OR honest pending_device; toolchain-provenance recorded UNCONFIRMED: until an Aurora device is in hand (§11.4.6)	E2E, FA, CHAL, REC	M

4. E23 — WASM browser-scoped MASQUE proxy (G23)

Browser-scoped proxy of the browser's own traffic — NOT a system-wide tunnel; the reuse pillar: helix-transport → wasm32 over WebTransport.

HVPN-P3-230 — helix-transport wasm32 build + WebTransport binding
(09–... §8; L)

id	Subtask	Acceptance	Tests	Cx
.1	Feature-gate helix-transport to compile to wasm32-unknown-unknown; bind browser WebTransport for the H3/datagram path (same MASQUE state machine, only the I/O leaf differs)	.wasm loads in headless Chromium; a MASQUE CONNECT-UDP session establishes to a test edge	UNIT,INT	L
.2	Throughput benched	proxy MB/s captured (BENCH)	BENCH	S

HVPN-P3-231 — Service-worker / fetch-shim integration + scope isolation
(09-... §8; M)

id	Subtask	Acceptance	Tests	Cx
.1	Service worker routing <i>only</i> explicitly-proxied origins through MASQUE; everything else normal browser networking	a non-proxied origin never touches the gateway (SEC, captured)	UNIT,INT,E2E	M
.2	Hard scope: only policy-authorized hosts (server-side need-to-know); die on tab/worker close	a non-authorized host is unreachable; tab-close terminates the session, no lingering tunnel (SEC)	SEC	M

HVPN-P3-232 — Console UX: explicit "browser-scoped, not a system VPN" affordance
(09-... §8; S)

id	Subtask	Acceptance	Tests	Cx
.1	ShieldIndicator variant + copy stating the limitation plainly; per-network toggle; live reachability	UX MP4 vision-verified that the scope wording is present + unambiguous (§11.4.159) — guards an over-claim bluff	UI,UX,REC	S

HVPN-P3-233 — G23 certification + threat note (WASM origin model)
(09-... §8; S)

id	Subtask	Acceptance	Tests	Cx
.1	Bank: browser reaches an authorized host, denied an unauthorized one, leaks nothing beyond scope; threat note (short-lived token, no WG private key in browser)	gate outcome=pass with captured evidence; the "no key in browser" invariant proven by a SEC assertion	E2E,SEC,CHAL	S

5. E24 — Billing-optional multi-tenant (G24)

Billing must not break no-logging — metering is aggregate counters only; billing OFF by default = zero behaviour change.

HVPN-P3-240 — Billing schema (plans/subs/entitlements/aggregate meters, RLS + no-log lint) (09–... §9; L)

id	Subtask	Acceptance	Tests	Cx
.1	FORCE-RLS tenant-scoped billing tables; usage_counter daily-aggregate only (no src/dst/ host/port/flow, no sub-day ts)	cross-tenant read denied (RLS); aggregate schema passes the lint	UNIT,INT,SEC	M
.2	Extend the no- log CI lint to reject a billing column matching the traffic-log shape + paired §1.1 mutation	the lint FAILs a planted usage_flows(src,dst,port,ts) table, passes the aggregate schema	SEC	S

HVPN-P3-241 — Aggregate metering pipeline (edge counters → daily upsert) (09–... §9; M)

id	Subtask	Acceptance	Tests	Cx
.1	Metering consumer folds edge rx/tx counters into usage_counter daily upserts (idempotent by tenant/device/day)	exactly-once bucket under a SIGKILL mid- fold (CHAOS)	UNIT,INT,CHAOS	M
.2	Bounded-memory at scale	10k devices, bounded memory (STRESS)	STRESS	S

HVPN-P3-242 — Entitlement enforcement in the coordinator (09–... §9; M)

id	Subtask	Acceptance	Tests	Cx
.1	Plan limits → admission checks at enroll + map build; over-quota = deny with reason (D-P3-2 fail- open)	an over-device-quota enroll is denied with a clear reason; bytes-quota breach degrades per the chosen policy	UNIT,INT,SEC	M
.2	PERF: check adds < 1 ms to map build	captured map-build delta < 1 ms	PERF	S

HVPN-P3-243 — Optional payment-provider adapter (interface + one reference impl, flagged) (09–... §9; M)

id	Subtask	Acceptance	Tests	Cx
.1	Narrow PaymentProvider interface + one reference adapter (provider behind a flag, no name in core §11.4.28)	billing-OFF path never calls the adapter (captured)	UNIT,INT,SEC	M
.2	Public webhook: signature-verify + replay-reject + rate-limit (first public surface → DDOS applicable)	webhook signature verified; replay rejected; rate-limit holds under flood (DDOS)	DDOS,SEC	S

HVPN-P3-244 — G24 certification (billing ON works, OFF is a perfect no-op, no-log preserved) (09–... §9; S)

id	Subtask	Acceptance	Tests	Cx
.1	Bank: (a) billing-ON meter+quota+payment flow; (b) billing-OFF byte-identical to pre-E24 (§11.4.108 runtime-signature diff); (c) no-log lint still green + planted traffic-log still FAILs	gate outcome=pass; the no-op proof captured as a §11.4.135 regression guard	E2E,SEC,FA,CHAL	S

6. E25 — Third-party security audit (G25)

A self-audit is not an audit; the verdict is the external party's. Engagement is operator-gated (spend → §11.4.101/.66).

HVPN-P3-250 — Audit scope, threat model, asset inventory (09–... §10; M)

id	Subtask	Acceptance	Tests	Cx
.1	STRIDE per surface (crypto core, control plane, the four reach surfaces, no-log invariant)	every shipped surface mapped to ≥ 1 threat + ≥ 1 audit objective; no surface unlisted (§11.4.118)	SEC	M

HVPN-P3-251 — SBOM + reproducible-source snapshot for the auditor (09–... §10; S)

id	Subtask	Acceptance	Tests	Cx
.1	CycloneDX SBOM per artefact + a	SBOM covers 100% of shipped deps; snapshot SHA recorded; endor/aikido-class	SEC,REPRO	S

id	Subtask	Acceptance	Tests	Cx
	pinned rebuildable source snapshot	scan green or every finding triaged		

HVPN-P3-252 — Engage auditor + run the engagement (operator-gated) (09–... §10; L)

id	Subtask	Acceptance	Tests	Cx
.1	Select + contract an independent firm (operator decision, §11.4.66 AskUserQuestion), share scope+snapshot, receive the report	report received; status Operator-blocked with the §11.4.21 unblock detail until the engagement is funded — honest, not faked	SEC,CHAL	L

HVPN-P3-253 — Remediation loop to zero Critical/High (§11.4.134/.146) (09–... §10; L)

id	Subtask	Acceptance	Tests	Cx
.1	Per finding: §11.4.102 root cause → §11.4.115 RED-on-vulnerable → fix → §11.4.146 extend → permanent §11.4.135 guard	zero unresolved Critical/High; each closure carries a falsifiable regression test (no console-mark-is-fixed bluff)	UNIT,INT,SEC,CHAL,REC	L
.2	Re-audit fixed items until the firm signs off (§11.4.134 iterate-to-GO)	the firm signs off after re-audit	SEC,CHAL	M

HVPN-P3-254 — G25 certification + public disclosure (09–... §10; S)

id	Subtask	Acceptance	Tests	Cx
.1	Publish the report + remediation summary; gate verdict	gate outcome=pass only when Critical/High = 0 + report published; else honest fail/operator_blocked with the open-finding list	SEC,CHAL	S

7. E26 — Reproducible builds (G26)

Table stakes for a privacy product: a user must verify the binaries match the source.

HVPN-P3-260 — Determinize Go + Rust builds (09–... §11; M)

id	Subtask	Acceptance	Tests	Cx
.1	Pin toolchains, <code>-trimpath/--remap-path-prefix</code> , fix <code>SOURCE_DATE_EPOCH</code> , strip non-determinism, build in a pinned containers image	two clean rebuilds of <code>helix-go+helix-edge+helix-core</code> on two hosts → identical SHA-256 (REPRO)	REPRO,FA	M

HVPN-P3-261 — Determinize the Flutter app builds (09–... §11; M)

id	Subtask	Acceptance	Tests	Cx
.1	Reproducible Flutter AOT (iOS/Android/desktop/web) + fork HAP/RPM; deterministic asset + ARB ordering	two rebuilds of each flavor → identical artefact (signature stripped before compare); divergences root-caused (§11.4.102) to zero	REPRO,FA	M
.2	Honest boundary: signed installers differ by signature → compare the unsigned payload digest	the honest boundary recorded (§11.4.6); unsigned-payload digests identical	REPRO	S

HVPN-P3-262 — WASM + native-shim reproducibility (09–... §11; M)

id	Subtask	Acceptance	Tests	Cx
.1	Pin OHOS/Aurora cross-toolchains; rebuild <code>.wasm</code> + <code>libhelix_ohos.so</code> + Aurora <code>libhelix.a</code>	two rebuilds → identical SHA-256, OR <code>PENDING_TOOLCHAIN</code> : honestly where a fork toolchain is not yet deterministic (a finding, not a pass)	REPRO	M

HVPN-P3-263 — Public verification script + transparency doc (09–... §11; S)

id	Subtask	Acceptance	Tests	Cx
.1	<code>verify-reproducible.sh</code> + <code>reproducible_builds.md</code> (model + honest signature exceptions)	a third party (the §11.4.165 independent verifier) runs it + confirms identity on every non-signature artefact (captured)	REPRO,FA,CHAL	S

HVPN-P3-264 — G26 certification (09–... §11; XS)

id	Subtask	Acceptance	Tests	Cx
.1	Gate verdict	gate outcome=pass only when every non-signature artefact	REPRO,FA,CHAL	XS

id	Subtask	Acceptance	Tests	Cx
	over the full artefact matrix	reproduces bit-identically on two independent hosts; any PENDING_TOOLCHAIN keeps the gate honestly open		

8. E27 — Reach l10n, governance, release

HVPN-P3-270 — First-tier ru + zh-Hans l10n completeness + overflow audit
(09-... §12; M)

id	Subtask	Acceptance	Tests	Cx
.1	Complete helix_l10n ARBs for ru/zh-Hans; no overflow/overlap (§11.4.162); state announced not just colored (ally, §04)	golden UI green in both locales on Access/Connector/Console; UX MP4 vision-verified	UI,UX,REC	M

HVPN-P3-271 — Phase-3 feature Status + Status_Summary set (§11.4.153) with per-surface video confirmation (09-... §12; M)

id	Subtask	Acceptance	Tests	Cx
.1	docs/features/Status.md + Status_Summary.md enumerating every Phase-3 surface (Implementation/Wiring/Validation/Video-confirmation), four-format export (HTML+PDF+DOCX), docs-chain-synced	every confirmed row has a real-use MP4 (PENDING_DEVICE rows honestly marked); §11.4.86 fingerprint matches	REC,CHAL	M

HVPN-P3-272 — docs-chain sync + items-DB load + reproducible-release tag
(09-... §12; S)

id	Subtask	Acceptance	Tests	Cx
.1	Load all HVPN-P3-* rows into docs/workable_items.db (§11.4.93/.95), sync exports (§11.4.65), cut <prefix>-3.0.0-... (§11.4.151) + fan out via merge-onto-latest-main (§11.4.113, never force-push)	workable-items validate green; tag prefix correct across main + every owned submodule; artefacts reproduce (E26)	FA,REPRO	S

9. Subtask roll-up + honest-gap ledger

Epic	Parent tasks	Deepened subtasks	Honest-gap class
E20 reach CI fabric	4	7	toolchain access (medium)
E21 HarmonyOS NEXT	5	11	PENDING_DEVICE — make-or-break
E22 Aurora OS	4	6	PENDING_DEVICE + Russian toolchain UNCONFIRMED
E23 WASM proxy	4	6	browser API maturity (medium)
E24 billing-optional	5	9	no-log invariant must hold
E25 third-party audit	5	7	Operator-blocked — external spend
E26 reproducible builds	5	7	fork-toolchain PENDING_TOOLCHAIN
E27 reach l10n/release	3	3	low
Total	35	56	—

Honest-gap ledger (§11.4.6, 09–... §13/§14 risk register). Phase 3 is the phase where some acceptances cannot be self-proven; the deepening preserves that honesty in the subtask acceptances rather than hiding it:

Subtask	Gap	Honest marker	Unblock (§11.4.148 D3)
211.2/.3, 214.1 HarmonyOS device reach	no HarmonyOS NEXT device/CI	PENDING_DEVICE: / gate pending_device	provision a device + add to the §11.4.128 tracked set (via G20)
221.2, 223.1 Aurora device reach	no Aurora device + Russian toolchain	PENDING_DEVICE: + UNCONFIRMED: provenance	provision an Aurora device; mirror-pin + audit the SDK
252.1 audit engagement	external spend	Operator-blocked (§11.4.101/.66)	fund + contract an independent audit firm
262.1, 264.1 fork-toolchain reproducibility	fork toolchain not yet deterministic	PENDING_TOOLCHAIN: keeps G26 honestly open	root-cause the fork non-determinism to zero

None of these may be marked pass on metadata/config-only evidence (09-... §13). A device-/engagement-gated gate that cannot be proven is pending_device/ operator_blocked with its unblock condition — the release notes state plainly which reach targets are *certified* vs *built-but-pending-device*. Complexity sums to $\approx$ 09-... §14's $\sim$ 172 engineer-days — the widest error bars in the programme, sizing only, never a date (§11.4.6).

Sources verified

- 09-phase3-reach-wbs.md §2.1 gates G20–G26, §5–§12 (every task Desc/Deliverable/Acceptance/Effort/Tests), §13 traceability + honest-gap rule, §14 risk register, §15 open decisions D-P3-* — read 2026-06-26.
- Sibling workable-items-model.md (§3/§6/§9 incl. gates table + pending_device/operator_blocked outcomes), dependency-graph.md (§6 Phase-3 DAG), subtask-deepening-p1.md/–p2.md (§0 conventions) — authored this volume.
- Constitution anchors
§11.4.3/.6/.10/.21/.27/.28/.40/.54/.66/.76/.86/.91/.93/.101/.102/.107/.108/.113/.115/.118/.128/.134/.135/.146/.151/.153/.156/.159/.161/.162/.165/.169 — read 2026-06-26.

Honest boundary (§11.4.6): subtasks decompose stated Phase-3 work. Device- and engagement-gated subtasks carry PENDING_DEVICE: / UNCONFIRMED: / Operator-blocked markers verbatim — a gate is never pass on metadata-only evidence. All complexity is sizing TARGET, never a date; Phase-3 carries the programme's widest error bars.